

PASSWORD STRENGTH CHECKER

PROJECT REPORT

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Project Title: Password Strength Checker

Technology Used: Python, CustomTkinter

1. ABSTRACT

Passwords are the first line of defense in protecting digital information and online accounts. Weak passwords can be easily guessed or cracked using brute force and dictionary attacks, leading to unauthorized access and security breaches.

The Password Strength Checker is a cybersecurity-based desktop application developed using Python and CustomTkinter. The application evaluates the strength of user-entered passwords by analyzing various security parameters such as length, uppercase letters, lowercase letters, numbers, and special characters.

The system provides real-time feedback, entropy calculation, strength classification, and suggestions for improving password security. The project aims to educate users about creating strong and secure passwords to enhance cybersecurity awareness.

2. INTRODUCTION

With the increasing use of online services, securing user accounts has become extremely important. Many users create simple and predictable passwords, making them vulnerable to cyberattacks.

A Password Strength Checker helps users determine whether their password is strong enough to resist common password-cracking techniques.

This project provides a user-friendly graphical interface that instantly evaluates password quality and offers recommendations for creating stronger passwords.

3. OBJECTIVES

The main objectives of this project are:

- To analyze password strength in real-time.
 - To identify weak passwords.
 - To provide security recommendations.
 - To calculate password entropy.
 - To educate users about secure password creation.
 - To improve cybersecurity awareness.
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4. PROBLEM STATEMENT

Many users create weak passwords such as:

- 123456
- password
- admin
- qwerty

These passwords are highly vulnerable to brute-force and dictionary attacks.

The challenge is to develop a system that evaluates password security and guides users in creating stronger passwords.

5. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 100 MB free space
- Display: 1366 × 768 resolution

Software Requirements

- Operating System: Windows / Linux
 - Python 3.10+
 - CustomTkinter Library
 - VS Code / PyCharm
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6. TOOLS AND TECHNOLOGIES USED

Python

Python is a high-level programming language used to implement the password analysis logic.

Advantages

- Easy to learn
 - Simple syntax
 - Large library support
 - Cross-platform compatibility
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CustomTkinter

CustomTkinter is a modern GUI framework built on Tkinter.

Features

- Modern UI components
 - Dark theme support
 - Custom styling
 - Responsive design
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Regular Expressions (Regex)

Regex is used to validate:

- Uppercase letters
 - Lowercase letters
 - Numbers
 - Special symbols
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Math Module

Used for entropy calculation.

7. METHODOLOGY

The system follows the following process:

Step 1

User enters a password.

Step 2

The application checks:

- Length ≥ 8
- Length ≥ 12
- Uppercase letters
- Lowercase letters
- Numbers
- Special characters

Step 3

The system calculates a security score.

Step 4

Password entropy is calculated.

Step 5

Strength level is displayed:

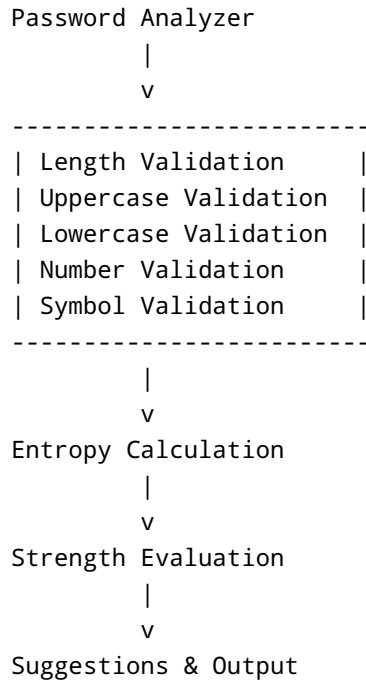
- Weak
- Medium
- Strong

Step 6

Suggestions are generated to improve password security.

8. SYSTEM ARCHITECTURE

```
User Input Password
      |
      v
```



9. PASSWORD STRENGTH PARAMETERS

The application checks:

Parameter	Description
Length	Minimum 8 characters
Uppercase	A-Z
Lowercase	a-z
Numbers	0-9
Symbols	@, #, \$, %, etc.

10. ENTROPY CALCULATION

Entropy measures password randomness.

Formula:

Entropy = Length × $\log_2$ (Character Set Size)

Higher entropy indicates stronger passwords.

Examples

Password	Entropy
Hello	28.5 bits
Hello123	47.63 bits
Hello@123	58.99 bits

11. STRENGTH CLASSIFICATION

Score Range	Strength
0 – 39	Weak
40 – 79	Medium
80 – 100	Strong

12. FEATURES IMPLEMENTED

Password Strength Analysis

Analyzes password complexity.

Real-Time Validation

Updates results while typing.

Entropy Calculation

Measures password randomness.

Show/Hide Password

Allows users to view entered passwords.

Security Checklist

Displays validation status.

Security Suggestions

Provides improvement recommendations.

Common Password Detection

Detects commonly used passwords.

Color-Based Strength Meter

- Red → Weak
- Yellow → Medium
- Green → Strong

Modern Cybersecurity Interface

Dark blue professional UI.

13. SAMPLE TEST CASES

Test Case 1

Password:

Hello

Output:

Weak

Entropy:

28.5 bits

Test Case 2

Password:

Hello123

Output:

Medium

Entropy:

47.63 bits

Test Case 3

Password:

Hello@123

Output:

Strong

Entropy:

58.99 bits

14. ADVANTAGES

- Easy to use
 - Fast analysis
 - User-friendly interface
 - Improves password security awareness
 - Real-time feedback
 - Lightweight application
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15. LIMITATIONS

- Does not check online password breach databases.
 - Does not store password history.
 - Limited to local password analysis.
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16. FUTURE ENHANCEMENTS

Future improvements may include:

- Password Generator
- Breached Password Detection

- Password History Tracking
 - PDF Report Generation
 - Multi-language Support
 - AI-based Password Suggestions
 - Cloud Integration
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17. APPLICATIONS

- Cybersecurity Awareness Programs
 - Educational Institutions
 - User Account Security
 - Personal Password Management
 - Security Training Sessions
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18. CONCLUSION

The Password Strength Checker successfully evaluates password security by analyzing password complexity, entropy, and character diversity. The application provides real-time feedback and useful recommendations that help users create stronger passwords.

The project demonstrates the practical application of cybersecurity concepts and secure password management techniques. It serves as an effective educational tool for promoting better password practices and improving overall digital security.

19. REFERENCES

1. Python Official Documentation
2. CustomTkinter Documentation
3. OWASP Password Security Guidelines
4. NIST Digital Identity Guidelines
5. Cybersecurity Best Practices Documentation